MediDesk - Clinic Appointment Manager

Own-domain capstone app. GitHub -> Actions -> GHCR(SHA) -> Terraform VPC+EKS -> Helm -> Ingress/HP
Prometheus/Grafana + ArgoCD.

Project Overview

Clinics lose track of walk-ins/priorities. MediDesk: KPI cards, filters, modal, stats API. Books appointment in <30s.

Tech Stack

FastAPI+SQLAlchemy+Alembic | React+Vite+Nginx unprivileged | Postgres 16 | pytest | Docker non-root | Actions+Trivy+GHCR | Terraform+EKS | Helm+Ingress+HPA | Prometheus+Grafana+ArgoCD.

Architecture Diagram

Browser -> Ingress (/ frontend, /api backend) -> FastAPI -> Postgres. /metrics -> Prometheus -> Grafana. Pu
-> Actions -> GHCR:SHA -> Helm -> EKS. Terraform: VPC+EKS.

CI/CD Flow

push->main: pytest gate + npm build; build x2 -> trivy HIGH,CRITICAL fail=1 -> push SHA tags; deploy: helm upgrade --install with SHA.

Infrastructure (Terraform)

VPC 10.0.0.0/16 3AZ public+private NAT; EKS 1.30 t3.medium 1-3 nodes; ap-south-1. init/plan/apply/destroy
secrets in git.

Kubernetes Setup

ns medidesk, 2 replicas, ClusterIP, Ingress, HPA 2-5 @70%, postgres PVC, ServiceMonitor. kubectl get pods/svc/ingress/hpa; helm list.

Observability

/metrics http_requests_total + latency; Targets UP; Grafana: rate, p95, 5xx live under load-test.sh.

Security (Trivy)

Trivy scans both images per push, fails on HIGH/CRITICAL. Secrets only via Actions secrets.

Key Learnings & Next Steps

Gates stop bad images; SHA links prod to commit; IaC beats click-ops. Next: RDS, TLS, OIDC, alerts, staging